Bloodbound

Blood Donor and Donation Camp Coordination System

User Guide

with setup instructions

Course	Software Engineering (CT-365), Complex Computing Problem
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1. Introduction

Bloodbound connects people who need blood with people who can give it. Hospitals post urgent blood requests. The system alerts compatible, eligible donors in the same city, and donors can accept with one tap. Camp organizers publish donation camps that donors can book, and on the day they check donors in and record each donation. Administrators keep the platform safe.

This guide explains how to use every part of the application, role by role, and how to set it up on a computer. It is written for the people who will use Bloodbound (donors, hospital staff, camp organizers and administrators), for the evaluators who will try it, and for the team members who need to run it.

1.1 One website, one account

There are no separate portals. Everyone uses the same website and the same sign-in page. After signing in, Bloodbound opens the workspace that matches your role.

Role	Who they are	Can create their own account?
Donor	A person who donates blood	Yes
Hospital staff	Hospital staff, or a patient's relative or requester, who posts blood requests	Yes
Camp organizer	Runs blood donation camps	Yes
Admin	Moderates users and requests	No. Admin accounts are created by the team directly in the database

1.2 How blood requests work

1. **Request.** Hospital staff post the blood group, component, units and urgency.
2. **Match.** Compatible, eligible donors in the same city are alerted automatically.
3. **Donate.** A donor accepts and then receives the hospital details. The hospital marks the request fulfilled, and the donor's next eligible date is tracked.

2. Getting started

2.1 The public homepage

Anyone can open the homepage without signing in. At the top you see the most urgent open blood request, with the blood group, the number of units needed, the hospital and city, the urgency level, how many donors have confirmed and how much time is left. If no request is open, a friendly message is shown instead.

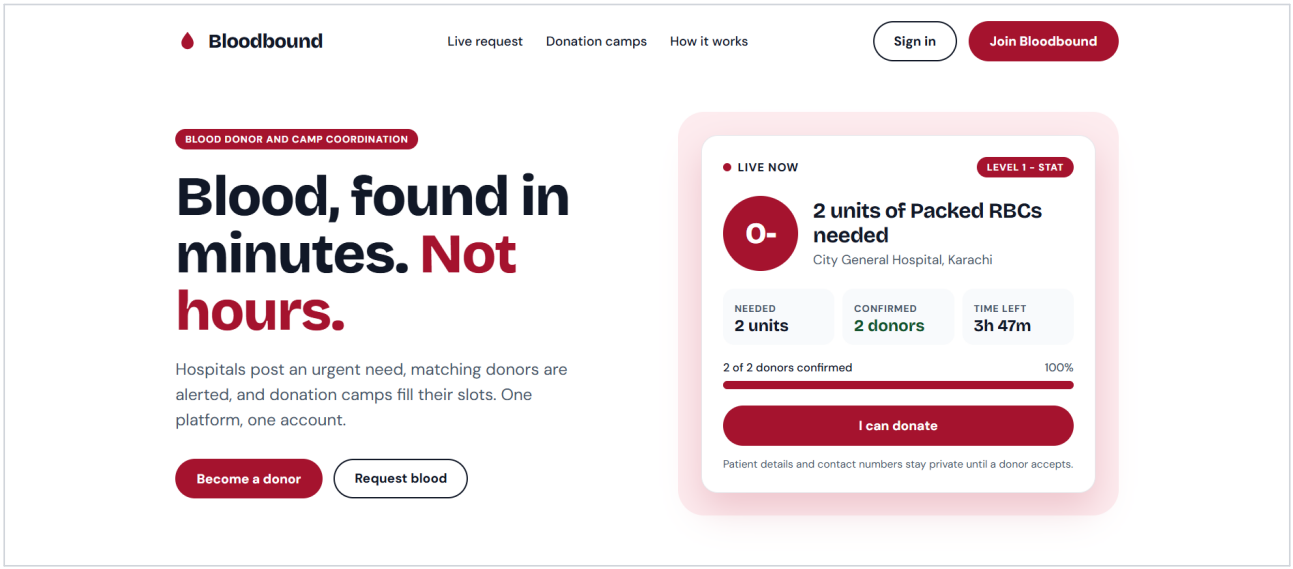


Figure 1. The homepage with the live blood request card.

> **Privacy**
> Patient details and contact numbers are never shown on the public page. They stay private until a donor accepts a request.

Scrolling down shows the eight blood groups and the donation camps that are open now, each with its venue, city, date, time and the number of slots left. Use **Sign in to book** to book a slot.

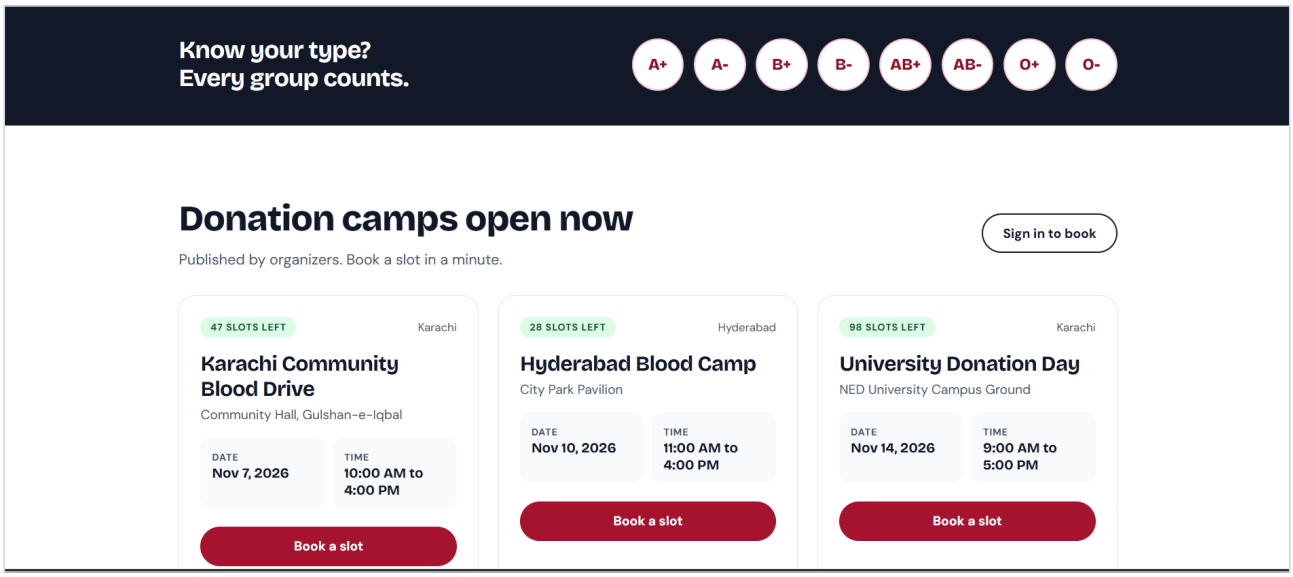


Figure 2. Blood groups strip and donation camps open now.

Further down, a short explanation shows the three steps (Request, Match, Donate), and organizers can follow **Create a camp** to start their own camp.

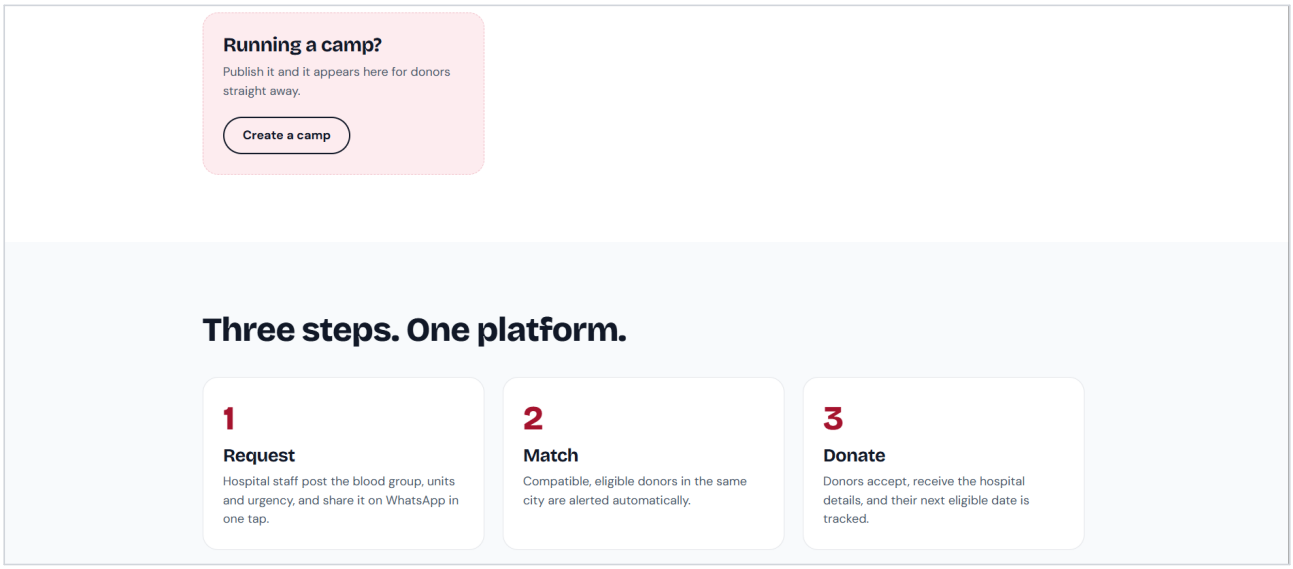


Figure 3. The "Running a camp?" call to action and the three-step explanation.

2.2 Creating an account

1. On the homepage choose **Join Bloodbound** or **Create an account**.
2. Fill in your name, email, phone, password and role.
3. If you are a donor, also choose your blood group and your city.
4. Submit the form, then sign in.

> Choose your role carefully. The role decides which workspace you see. Hospital staff, camp organizers and donors can register themselves. Admin is not available on the form.

2.3 Signing in

Enter your email and password in the **Welcome back** box and press **Sign in**. Bloodbound takes you straight to the workspace for your role. A sign-in stays valid for 24 hours, then you must sign in again.

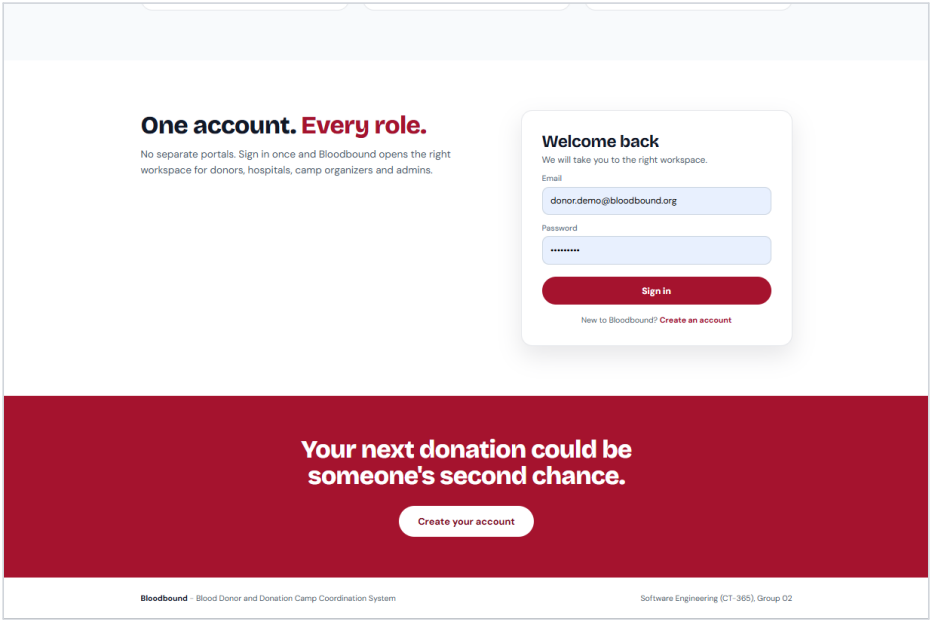


Figure 4. The sign-in form on the homepage.

2.4 Moving around

A navigation bar at the bottom of the screen shows the sections available to your role. The top bar shows your name and a **Log out** button. Always log out when you use a shared computer.

3. Donor guide

Navigation: **Home, Alerts, Camps, Profile**.

3.1 Home

Your home screen shows whether you are eligible to donate, a reminder message with your next eligible date, your last donation date, and your upcoming camp bookings. A banner appears when emergency alerts are waiting for you, with a button that takes you straight to them.

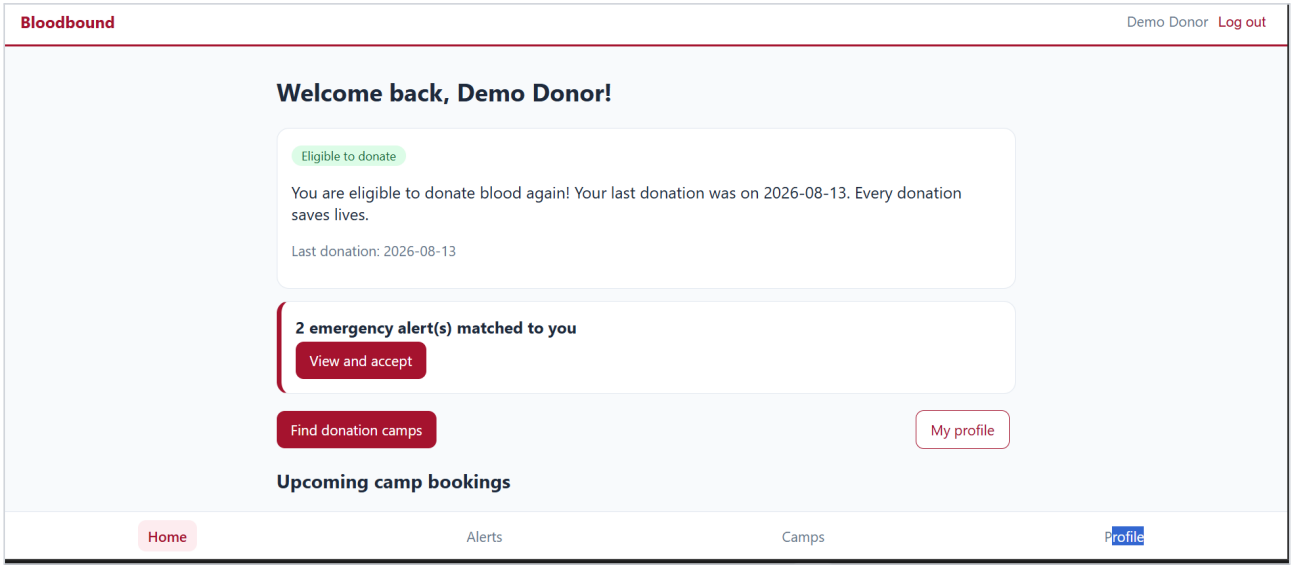


Figure 5. Donor home: eligibility, alert banner and shortcuts.

3.2 Understanding eligibility

- After you donate, you must wait before donating again. The waiting period is 56 days by default and can be configured by the team.
- Your eligibility is worked out from your last donation date, and the app shows your next eligible date.
- While you are not eligible, you are not matched to new requests, and you cannot accept requests or book camp slots.

3.3 Alerts: accepting an emergency request

The **Alerts** page lists emergency requests that match you: your city, a compatible blood group, and you are eligible. Each alert shows the blood group, component, units needed, urgency, hospital, city and when it expires.

1. Open **Alerts** and read the request.
2. Press **Accept** if you can donate.
3. After accepting, the hospital address and contact number appear. Go to the hospital.
4. If your plans change, press **Withdraw** so that another donor can take your place.

> **Why contact details are hidden**
> The hospital's address and phone number are hidden until you accept. Your name and phone number are visible to the hospital only after you accept.

3.4 Camps: booking a slot

1. Open **Camps**. Use the city filter to narrow the list.
2. Check the slots left, then press **Book slot**.
3. Your booking appears under **My bookings** and on your home screen.
4. To cancel, press **Cancel booking**. Only future bookings that you have not yet attended can be cancelled.

You can book a camp only once, and you cannot book a camp that is full or not yet published. The slot must fall inside the camp hours and your eligibility is checked for the date of the camp.

3.5 Profile

The **Profile** page shows your eligibility and lets you update your blood group, city and last donation date. Keep your city correct, because matching is done by city. If you donated somewhere else, update your last donation date so your eligibility stays accurate.

4. Hospital staff guide

Navigation: **Home, New request, My requests**.

4.1 Home

The home screen shows how many requests you have in total, how many are open and how many are fulfilled, with shortcuts to create a request or open your list.

4.2 Creating a blood request

Open **New request** and fill in the form. When you submit, matching donors are alerted immediately and you see how many were alerted.

Field	What to enter
Patient identifier	A reference for the patient (not shown to donors or on the public page)
Blood group	A+, A-, B+, B-, AB+, AB-, O+ or O-
Component	Whole Blood, Packed RBCs, Platelets or Plasma
Units	From 1 to 10
Urgency	Critical, High or Normal (see the table below)
City	The city where donors should be found
Hospital name and address	Shown to a donor only after they accept

Urgency and automatic expiry

Urgency	Level	Request expires after
Critical	Level 1 (Stat)	4 hours
High	Level 2 (Urgent)	12 hours
Normal	Routine	24 hours

4.3 Following your requests

My requests lists each request with its status and urgency, and how many donors were alerted and how many accepted. Open a request to see its details.

4.4 Request detail

- **Accepted donors:** their names and phone numbers appear here once they accept, so you can coordinate the donation.
- **Share on WhatsApp:** opens a ready-written WhatsApp message that you can send to groups. Nothing is sent automatically. The message describes the need and leaves out the patient identifier.
- **Mark fulfilled:** available once at least one donor has accepted. It closes the request and updates each accepted donor's last donation date.
- **Cancel request:** closes a request that is no longer needed.

4.5 Request statuses

Status	Meaning
Pending	Open. No donor has accepted yet.
Matching	At least one donor has accepted.
Fulfilled	You confirmed the donation.
Cancelled	Closed by you or by an administrator.
Expired	The time limit for the urgency level passed.

If enough donors have accepted to cover the units you asked for, further donors cannot accept.

5. Camp organizer guide

Navigation: **My camps, New camp**.

5.1 Creating a camp

Open **New camp** and enter the camp name, address, city, start and end time, maximum capacity and target units. A new camp is saved as a **Draft**, which donors cannot see.

Rule	Limit
Camp name	5 to 100 characters
Maximum capacity	10 to 1000 donors
Start time	Must be in the future
End time	Must be after the start time

5.2 Managing your camps

My camps shows each camp with its status, bookings, check-ins and units collected.

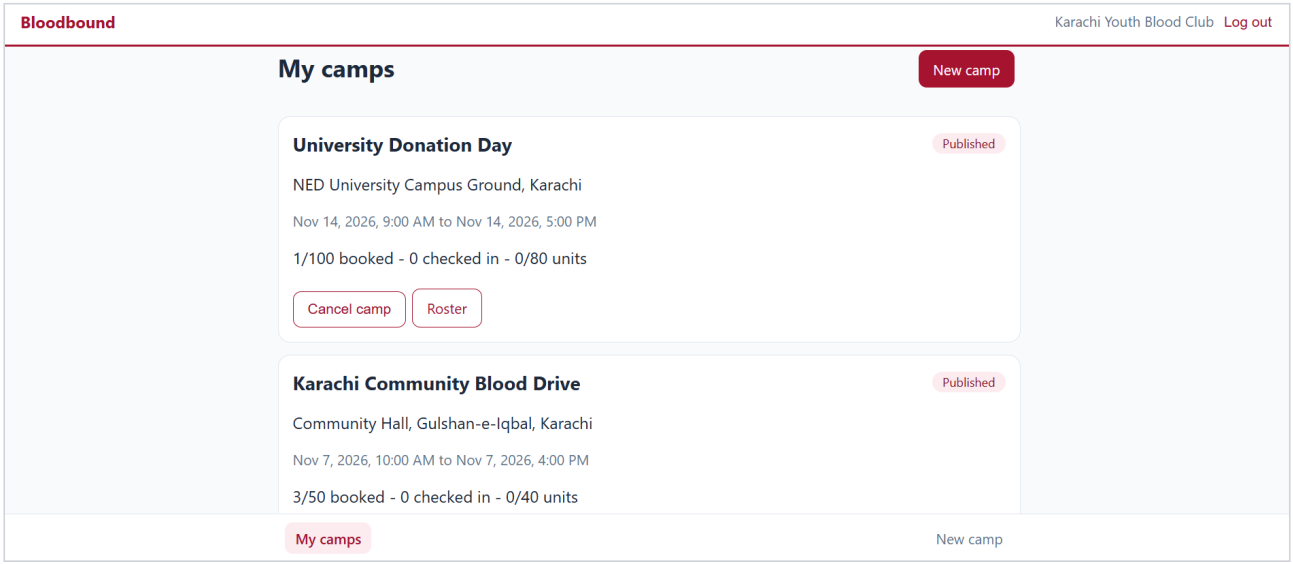


Figure 6. An organizer's camp list with status, bookings and actions.

- **Publish:** moves a Draft camp to Published. It then appears on the homepage and in donors' camp lists.
- **Cancel camp:** closes a Draft or Published camp.
- **Roster:** opens the list of donors who booked.

5.3 On the day: check-in and recording donations

1. Open the camp and press **Roster**.
2. When a donor arrives, press **Check in**.
3. After they donate, choose the blood component and press **Record donation**.

You must check a donor in before recording a donation, and only one donation can be recorded for each booking. Each recorded donation adds a unit to the blood inventory and to the camp's units collected, and sets the donor's last donation date to today.

5.4 Camp statuses

Status	Meaning
Draft	Being prepared. Not visible to donors.
Published	Open for booking.
Ongoing	The camp is taking place now. Still open for booking.
Completed	The camp has ended.
Cancelled	Closed by the organizer or by an administrator.

6. Administrator guide

Navigation: **Dashboard, Users, Requests**.

6.1 Dashboard

The dashboard gives a live overview: users (total, active, deactivated, and by role), blood requests by status, camps by status, and recent requests.

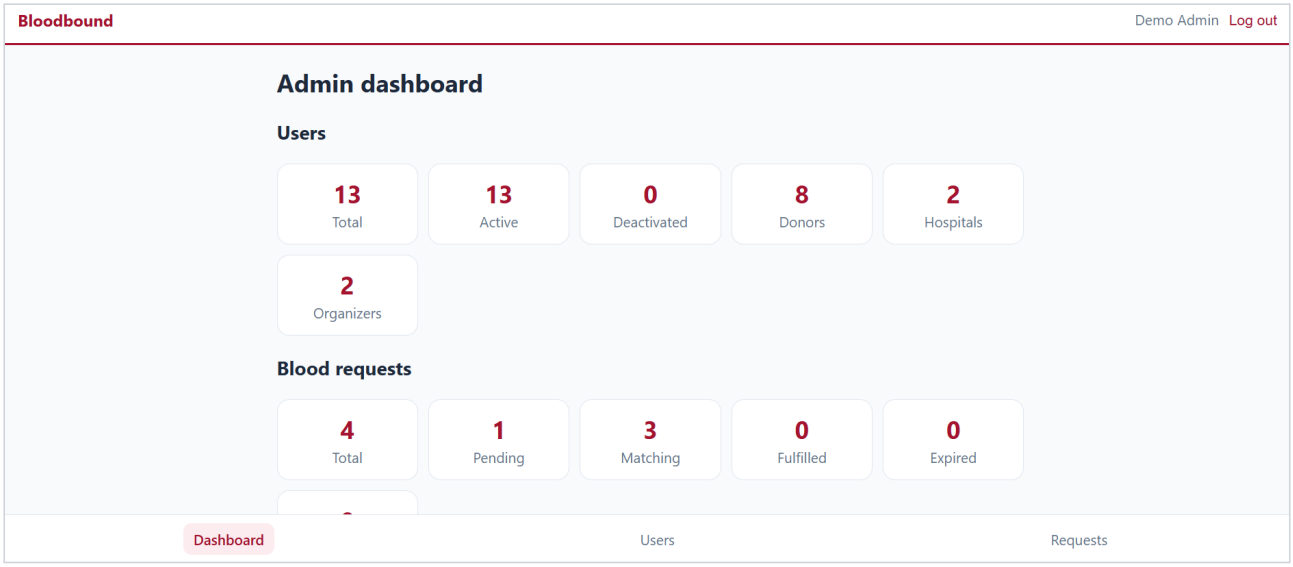


Figure 7. The admin dashboard.

6.2 Managing users

- Search for a user by name, email, phone or city.
- **Deactivate** a user you suspect of misuse. You are asked for a reason. A deactivated user cannot sign in.
- **Reactivate** a user to restore access. You cannot deactivate your own account.

6.3 Moderating requests

Requests lists every blood request from every hospital, with the requester and the number of acceptances. Open requests that look false or abusive can be **expired** immediately or **removed**, with a reason. This protects donors from fake requests.

> **Note**
> Camp monitoring exists in the system's API, but there is no admin camps screen in this version.

7. Privacy and safety

Bloodbound shares the minimum information needed, and only at the moment it is needed.

Information	Who can see it, and when
Patient identifier	Only the hospital that created the request. Never donors, the public page or WhatsApp messages.
Hospital address and phone	A donor sees them only after accepting that request.
Donor name and phone	The hospital sees them only after the donor accepts its request.
Public homepage	Shows blood group, units, hospital name, city, urgency and counts. No addresses, phone numbers, emails, coordinates or user IDs.
Passwords	Stored only as one-way hashes. Nobody, including administrators, can read them.
Sign-in sessions	Expire after 24 hours.

7.1 Safety rules built into the system

- Only compatible blood groups are matched, using standard red-cell rules. Plasma follows its own, reversed rules.
- A donor must be eligible to be matched, to accept, or to book a camp.
- Roles are enforced on the server. A donor cannot open hospital, organizer or admin features, even by typing an address.
- Administrators can deactivate abusive accounts and remove false requests.

8. Troubleshooting and frequently asked questions

I cannot sign in.

Check the email and password. If you signed in more than 24 hours ago, sign in again. If everything is correct, your account may have been deactivated by an administrator.

I see no alerts as a donor.

Alerts appear only when a request matches your city, your blood group is compatible and you are eligible. Check your eligibility on the home screen and your city on your profile. A request also disappears when it expires or when enough donors have already accepted.

I cannot accept a request.

The request may have expired, been cancelled or fulfilled, or already have enough accepted donors. You must also be eligible and matched to it.

I cannot book a camp slot.

The camp may be full, not published, or you may already have booked it. Your eligibility is checked for the camp date, so a recent donation can block booking a camp that takes place before your next eligible date.

The "Mark fulfilled" button is missing.

A request can be fulfilled only after at least one donor has accepted it.

My request disappeared from the live list.

It expired (4, 12 or 24 hours depending on urgency), or it was cancelled, fulfilled or removed by an administrator.

A donor cannot be checked in or recorded.

A donor must have booked the camp. Check in first, then record the donation. A booking can have only one donation.

I forgot my password.

This version has no password reset screen. Ask the team to reset it directly in the database.

Can I edit my details?

Donors can edit their blood group, city and last donation date on the profile page. Hospital, organizer and admin accounts cannot edit their profile details in this version.

Does WhatsApp sharing send messages automatically?

No. It opens WhatsApp with a ready-written message, and you choose who to send it to.

Does it work on a phone?

The layout adapts to smaller screens, but the application was designed and checked on a desktop browser.

Appendix A. Setup instructions

These steps run Bloodbound on your own computer.

A.1 What you need

- Node.js (a current LTS version) and npm
- Git
- A PostgreSQL database. The project uses a free hosted database on Neon, so nothing needs to be installed locally.
- A modern web browser

A.2 Get the code

```
git clone https://github.com/AreebaAliSiddiqui/BloodBound.git
```

A.3 Start the back end

1. Open a terminal in `SourceCode/backend` and run `npm install`.
2. Create a file named `.env` (copy `.env.example`) with: `DATABASE_URL` (your database connection string), `JWT_SECRET` (any long random text), and optionally `PORT` (default 5000) and `ELIGIBILITY_GAP_DAYS` (default 56). Never share or commit this file.
3. Create the database tables by running `database/schema.sql`, then every file in `database/migrations` in numeric order.
4. Start the server with `node server.js`. It listens on port 5000. You should see "Server is running on port 5000".

A.4 Load demo data (optional)

- `node scripts/seed-demo-users.js` creates one demo account for each role.
- `node scripts/reset-database.js --yes` followed by `node scripts/load-mock-data.js` resets the database and fills it with demo requests, camps, bookings and acceptances (from `InputData/mock-data.json`).

> **Warning**
> The reset script deletes all existing data. Check which database your `.env` points to before running it.

A.5 Start the front end

1. Open a second terminal in `SourceCode/frontend` and run `npm install`.
2. Run `npm run dev` and open the address that Vite prints (usually `http://localhost:5173`).
3. The front end calls the back end through `/api`, which Vite forwards to port 5000, so keep the back end running.

A.6 Run the automated tests

With the back end running, open another terminal in the project root and run:

```
node --test --test-concurrency=1 "Tests/*.test.js"
```

All tests should pass. Afterwards, remove the test data with `node Tests/cleanup.js`. Written test reports are in the `TestReports` folder.

Appendix B. Demo accounts

After loading the demo data, every account below uses the password **Demo@1234**. These accounts are for demonstration only.

Role	Email
Admin	admin.demo@bloodbound.org
Hospital staff	hospital.demo@bloodbound.org hyd.hospital.demo@bloodbound.org
Camp organizer	organizer.demo@bloodbound.org hyd.organizer.demo@bloodbound.org
Donor	donor.demo@bloodbound.org (O-, Karachi) ayesha.donor, sana.donor, hamza.donor, mehak.donor, usman.donor, zara.donor (all @bloodbound.org) bilal.donor@bloodbound.org (donated recently, so not eligible)

The demo data includes four blood requests (two in Karachi, two in Hyderabad), four camps (three published and one draft) and several bookings and acceptances.

Appendix C. Glossary

Term	Meaning
Eligible	Allowed to donate because the waiting period since the last donation has passed.
Component	The type of blood product: Whole Blood, Packed RBCs, Platelets or Plasma.
Matching	Finding donors who are in the same city, have a compatible blood group and are eligible.
Roster	The list of donors booked for a camp.
Check-in	Marking that a booked donor has arrived at the camp.
Inventory	The record of blood units collected at camps, with their expiry dates.
Deactivated	An account blocked from signing in by an administrator.
Stat (Level 1)	The most urgent request level, which expires after 4 hours.